Problem 18.1

$Ax = 0$ when entries of every row of A add to 0 is $x = [1\ 1\ 1\ \ldots\ 1\ 1]^T$. Since the nullspace is not the 0 vector alone, the matrix A is singular so $|A| = 0$.

If those entries of A add to 1, then those entries of $A - I$ add to 0 because the ~~rows~~ sum of the entries of ~~rows of A ge~~ of each row of A get 1 subtracted thus adding to 0 instead, and so by the same logic as the original A, $\det(A - I) = 0$.

$\det(A - I) = 0$ ~~implies det A = 1~~. ~~Th~~ does not imply $\det(A) = 1$.
For example, let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. ~~the~~

The $A -$ ~~has~~ I has determinant 0 but A has determinant -1.

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Problem 18.2

$$\begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & 0 & c^2-a^2-(b+a)(c-a) \end{bmatrix}$$

Matrix is in triangular form so determinant is:- $(b-a) \times ((c^2-a^2) - (b+a)(c-a))$

$= (b-a)((c+a)(c-a) - (b+a)(c-a))$

$= (b-a)(c-a)((c+a) - b - a)$

$= (b-a)(c-a)(c-b)$